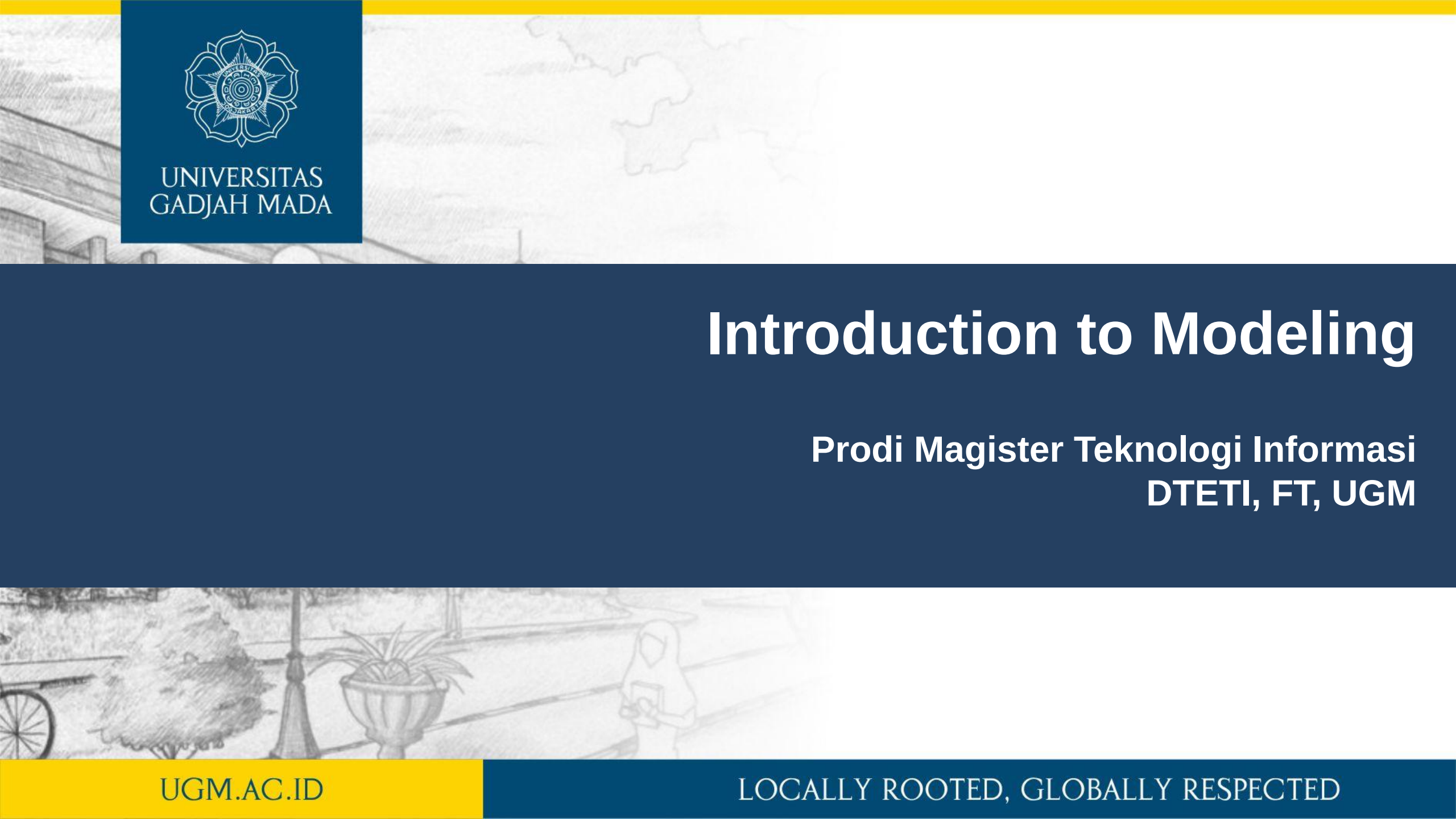


Introduction to Modeling

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Are problem-solving and modeling related?

What is problem-solving?

What is modeling?

What is modeling?

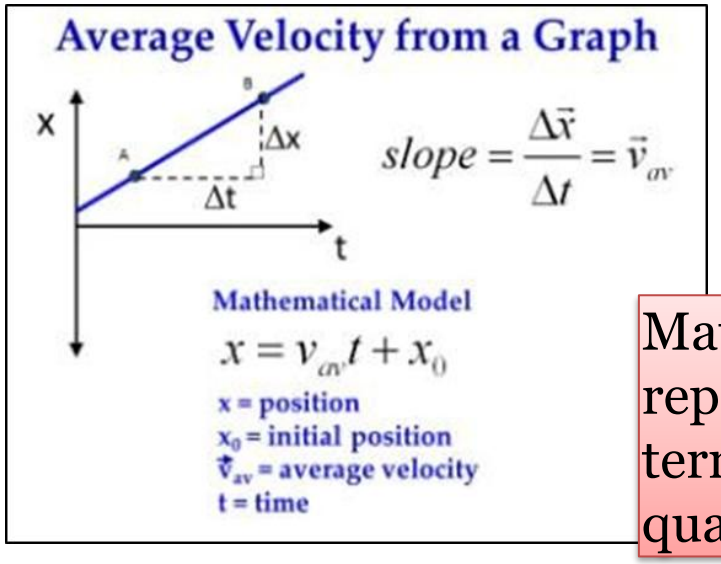
Modeling is the process of making/creating a model.

A model is

- A simpler representation of something (item, phenomenon, system, process).
- Models can also be said to be replicas; they can be physical or mathematical.







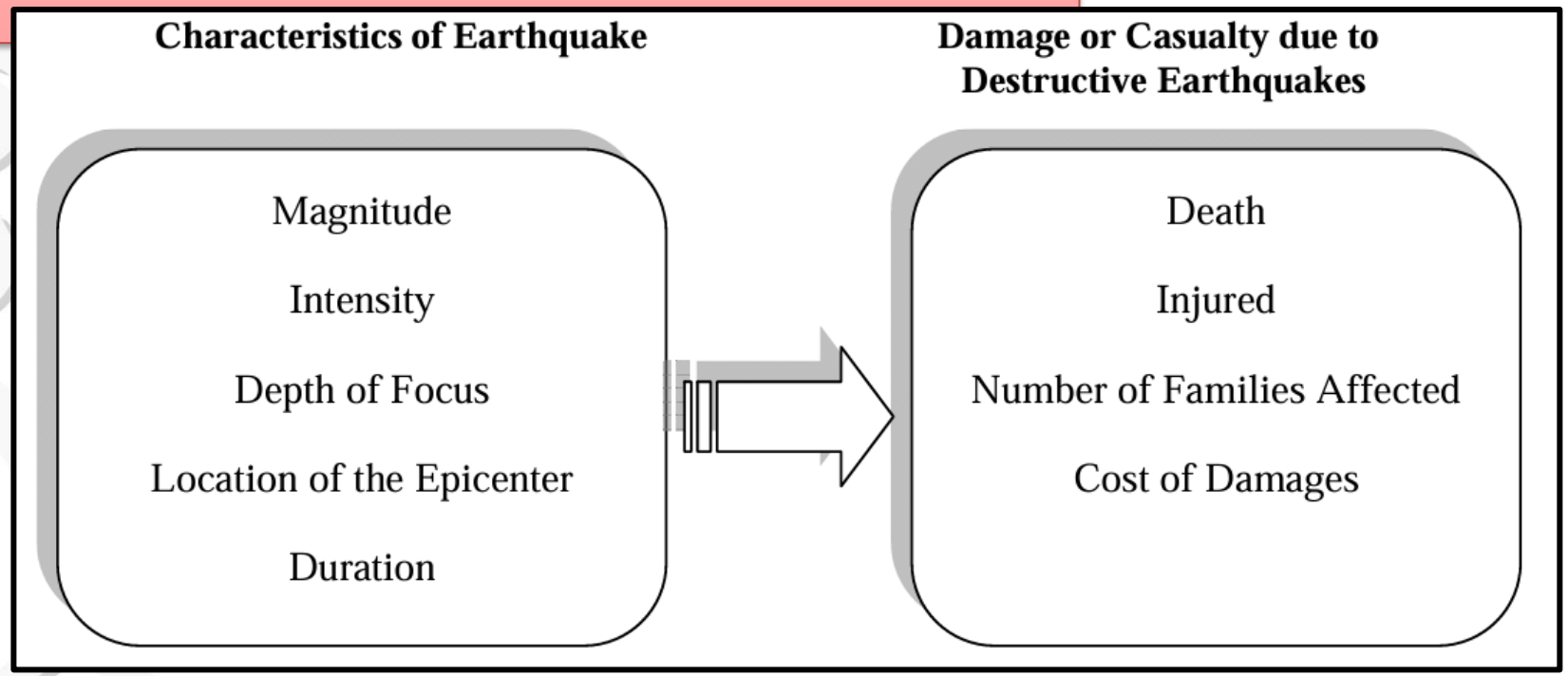
Mathematical modeling is representing a system in terms of logical and quantitative relationships.

Are problem-solving and modeling related?

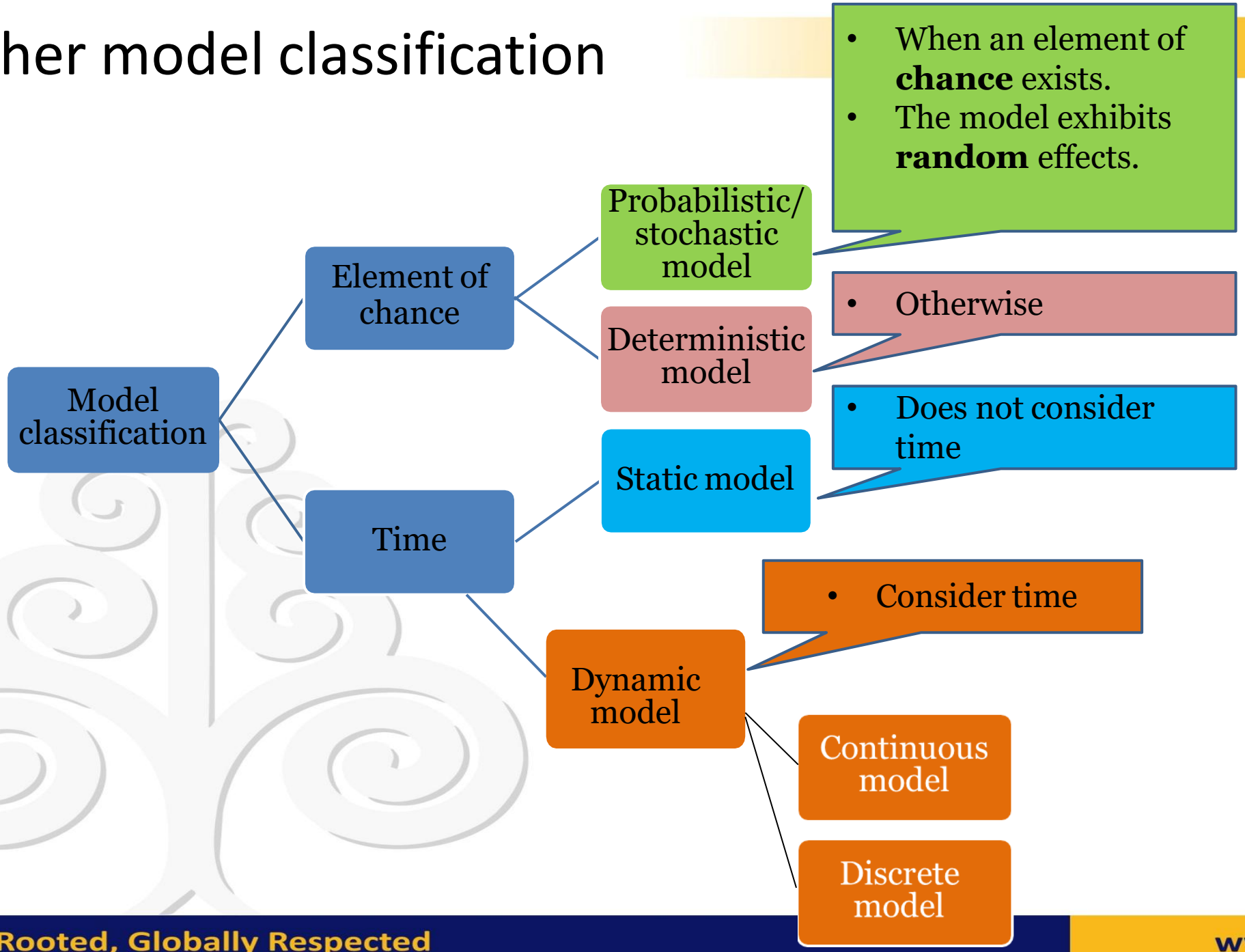
- We use modeling to understand the problem by creating simplified version of the real-life systems.

- We can use modeling to **analyze** complex, real-world problems to **predict** what might happen with some course of action.
- Example:
  - Use math models to simulate effects of earthquake (making quantitative predictions of infrastructural damages, response, and recovery to help minimize damage, death, and injury).



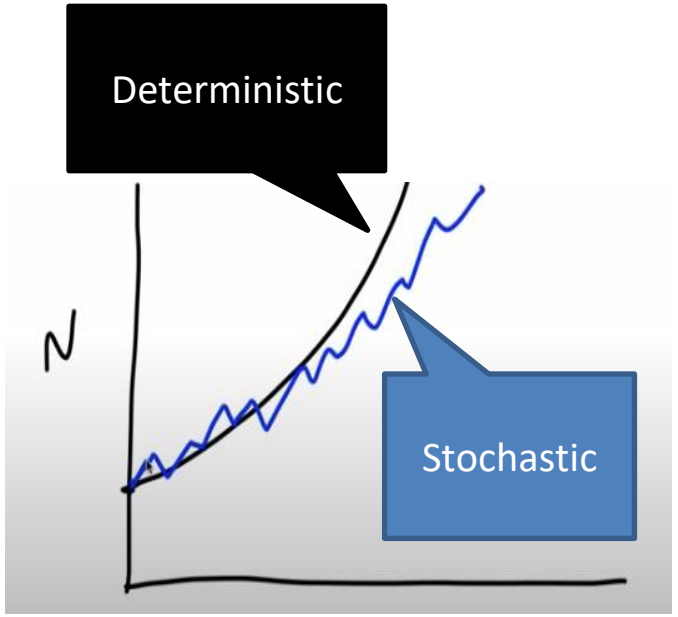


Another model classification



Examples of stochastic and deterministic models

- Stochastic models have elements of randomness.
  - The same parameter values and the same initial conditions can lead to different outcomes.
- Let's compare deterministic with stochastic:
  - If population growth could be modeled by
    - $N(t+1) = N(t)r$, where r is the growth rate → deterministic model
    - $N(t+1) = N(t)r(t)$, where r(t) indicates that the growth rate may change over time for some reason (random) → stochastic model



Static model represents of a system at a particular time, or one that may be used to represent a system in which time simply plays no role.

- Static (steady-state) models consider systems in equilibrium, thus are time-invariant.
- Static models are more structural than behavioral.
- Static modeling is more rigid than dynamic modeling because it is a time-independent view of the system.
- Example:
  - replica of sea or building or car
  - calculation of mechanical stress in a bridge

Contoh model dinamis

Dynamic model represents a system as it evolves over time, such as a conveyor system in a factory.

- Dynamic modeling is flexible because it can change over time because it shows what an object does with many possibilities that may arise in time.
- Example:
  - Math model of mechanical stress in a bridge that consider effects of **moving** traffic, wind, earthquakes, or other things that vary in time.

Continuous and discrete model

Definitions In a **continuous model**, time changes continuously, while in a **discrete model**, time changes in incremental steps.

- We often develop discrete computer models of continuous situations.
- Example?

Steps of Modeling Process - Cyclic

Analysis of the problem

Formulate model

- Gather data
- Make simplifying assumptions and document them
- Determine variables and unit
- Establish relationship between variables
- Determine equation and functions

Steps of Modeling Process - Cyclic

Solve the model

Verify and interpret the model solution

Steps of Modeling Process - Cyclic

Report on the model

- Analysis of the problem
- Model design
- Model solution
- Result and discussion

Maintain the model

Exercise

1. Compare and contrast the modeling process with the scientific method: Make observations; formulate a hypothesis; develop a testing method for the hypothesis; collect data for the test; using the data, test the hypothesis; accept or reject the hypothesis.
2. Compare and contrast the modeling process with the software life cycle: analysis, design, implementation, testing, documentation, maintenance.

What is simulation?

Imitation of the operation of a real-world process or system over time.

→ need to develop a simulation **model.**

- Examples: …

- The process of
  - designing a mathematical or logical model of a real-system and then
  - conducting computer-based experiments with the model to
    - describe, explain, and predict the behavior of the real system.
- The actual running of the model system to gain insight into its performance.
- The process which simulates in the laboratory or on the computer, the actual scenario as close to the system as possible.

Simulation benefit: predict the system behavior & suggest improvements in it

Use simulations if

- Not feasible to do actual experiments
- System does not exist (VM ware)
- Cost of actual experiments prohibitive (discourage)
  - Money
  - Time
  - Danger
- Want to test alternatives

Actual Trial vs Simulation

e.g. damage from weapon

- **Actual trial** (some says actual simulation)

  => drop the weapon on the target and see its end effects.

- **Computer/laboratory simulation**

  => conducting trials on computer/laboratory, which simulates the end effects of the weapon on the target.

Any questions?